

1 **DEAN'S POLICY AND OPERATING MEMORANDUM 03-18***

2
3 OFFICE OF THE DEAN
4 UNITED STATES MILITARY ACADEMY
5 WEST POINT, NEW YORK 10996-5000
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11 **ACADEMIC LABORATORY RESOURCE PLAN**

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35 1. **Purpose.** This document establishes the procedures and responsibilities for equipping and
36 supplying curriculum-supporting academic laboratory programs (hereafter called laboratory
37 programs) at the United States Military Academy (USMA). DPOM 03-18 also serves as a guide
38 for resource managers in academic departments to sustain these laboratory programs. DPOM
39 03-18 is intended to synchronize resource planning and execution efforts across departments
40 with suspense dates imposed by the resource allocation and execution system at USMA.
41

42 2. **Applicability.** This document applies to all departments at USMA with at least one
43 laboratory program supported by the Laboratory Equipment and Resources Committee (LERC).
44

45 3. **General.**

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47 a. **Laboratory Programs.** Each department participating in the LERC has at least one
48 laboratory program supported by the Academic Laboratory Resource Plan. These laboratory
49 programs represent distinct groupings within each department's overall laboratory structure.
50 Laboratory programs will support one or more courses offered by the department. A laboratory
51 equipment item is purchased to support one laboratory program but may support two or more

*This DPOM supersedes DPOM 03-18, dated 2 July 2008 and remains in effect until superseded.

laboratory programs. Departments assess the status of laboratory programs (see Para 5.a. Status Codes) based on an overall evaluation of the present laboratory program's ability to meet its objectives, funding allocation impacts on this level of capability, and the importance of any shortfall on the capability of that program. ENCLOSURE A lists the laboratory programs supported by the Dean's laboratory budget. Departments may not add laboratory programs without approval of the Dean. (See Para. 9 below for procedures.)

b. Funds allocated according to the procedures described in this academic laboratory resource plan are for supplies and equipment solely for the curriculum-supporting academic laboratory programs. Allocation of resources for laboratory supplies and equipment is the responsibility of the LERC. Laboratory supplies include items consumed in the execution of a laboratory exercise and items traditionally required for the day-to-day operation of laboratory equipment. Laboratory equipment supporting laboratory programs discussed in this memorandum includes:

- (1) laboratory equipment and instrumentation,
- (2) computers and software used for laboratory equipment control or data acquisition,
- and
- (3) computers and software supporting computer laboratories.

4. Allocation of Appropriated Funds.

a. Types of funds. Several types of appropriated funds are made available to the LERC to support laboratory programs:

(1) Operations & Maintenance – Army (OMA) Funds – Funds appropriated for one year to cover operating and maintenance costs in the year of appropriation. OMA funds can be used to pay for goods and services where individual items are less than the approved OMA/OPA threshold.¹

(2) Other Procurement – Army (OPA) – Funds appropriated for a three-year period to procure capital equipment. OPA funds must be used on individual items that exceed the approved OMA/OPA threshold. OPA funds must also be used to procure systems of equipment when the accumulated system cost exceeds the approved OMA/OPA threshold.

(a) OPA – 2 Funds – OPA funds to be used on Automated Data Processing (ADP) equipment or systems meeting the OPA threshold.

(b) OPA – 3 Funds – OPA funds to be used on non-ADP equipment or systems meeting the OPA threshold.

¹ As of 17 July 2024, the OMA/OPA threshold was \$350k. See: "H.R.2882 - 118th Congress (2023-2024): Further Consolidated Appropriations Act, 2024." [Congress.gov](https://www.congress.gov/bill/118th-congress/house-bill/2882), Library of Congress, 23 March 2024, <https://www.congress.gov/bill/118th-congress/house-bill/2882>. or "Text - S.2587 - 118th Congress (2023-2024): Department of Defense Appropriations Act, 2024." [Congress.gov](https://www.congress.gov/bill/118th-congress/senate-bill/2587/text), Library of Congress, 27 July 2023, <https://www.congress.gov/bill/118th-congress/senate-bill/2587/text>. or "2024 Fiscal Law Deskbook." The Judge Advocate General's Legal Center and School, United States Army, Charlottesville, Virginia, 2024.

b. Funds Allocation.

(1) OMA Funds Allocation. OMA funds are allocated to USMA at the beginning of the fiscal year, or about one month after Congress approves the federal budget. The Office of the Dean is allocated a portion of the USMA funds. The Office of the Dean determines how much OMA money will be allocated to laboratory requirements based on the funds available and overall requirements for the fiscal year. These OMA funds are then allocated to the departments based on prioritized laboratory program needs through the LERC fund allocation process. Funds are allocated to laboratory programs in two forms: supplies and maintenance (S&M) funds and replacement and upgrade (R&U) funds. The LERC makes prioritizing recommendations to the Dean when there are inadequate funds available to meet the needs of every laboratory program in each department.

(2) A list of Unfinanced Requirements (UFRs) is developed by the LERC, to address requirements that are not funded by the Dean's allocation of funds to the LERC. This list is submitted to the Vice Dean of Resources (VDR) for possible allocation of funds at mid-year or at the end of the fiscal year.

(3) OPA Funds Allocation. OPA funds are centrally controlled by the Army Staff. Departments with laboratory equipment items or systems qualifying for OPA funding will submit justifications, requisitions, and other required forms through the Assistant Dean for Information Technology (ADIT) and the Assistant Dean for Programs and Resources to the VDR. OPA-2 funding requests are submitted through ADIT and the Assistant Dean for Programs and Resources to the VDR.

5. Funds Allocation Process.

a. Status Codes. The status of each of the laboratory programs overseen by the LERC is reported using a system of color-codes. The codes reflect the extent to which a laboratory program is equipped to meet the educational objectives for the courses that it supports. Laboratory programs can be coded as Green, Amber, Red, or Black:

(1) GREEN: "fully capable." To be rated GREEN, a laboratory program must be equipped to efficiently support all educational objectives the laboratory program was designed to achieve.

(2) AMBER: "degraded capability." AMBER programs suffer from a lack of adequate equipment on hand and/or significant maintenance problems but remain no less than marginally capable of supporting specified educational objectives. AMBER laboratories generally promote inefficiency because they require additional staff and faculty time to operate. If several of a department's laboratory programs are AMBER, then the cumulative effect on overall efficiency can be greater than the sum of the parts since the same staff and faculty members often operate more than one laboratory. The direct impact on cadets includes a reduction in the breadth of the laboratory experience, a reduction in depth of the experience, or both.

(3) RED: "program pending failure." The RED status is designed to alert decision makers that unless decisive action is taken, the laboratory program will likely fail (i.e. become BLACK) within 6 – 12 months. A program would be classified as RED if a critical piece of equipment is on the verge of breaking down but is still operational. Another case for which the RED status is appropriate is when critical equipment is not present or operational but is not

required until the next term. To prevent a RED program from degrading to a BLACK status, funds must be provided in time for the necessary equipment to be procured before it is needed.

(4) BLACK: “program failure now.” BLACK programs are incapable of supporting desired educational objectives. Unless funding is immediately provided, large-scale changes to educational objectives must be made.

b. To determine levels of support and prioritize departmental requirements, laboratory equipment items are classified according to their impact on the Status Code of an individual laboratory program. A laboratory equipment item can be classified as MSL, CSL, or FSL.

(1) MSL: “Minimum Sustainment Level.” Laboratory equipment that if not funded will result in a laboratory program in RED or BLACK status. If funded this would allow a laboratory program to maintain AMBER or GREEN status through the current fiscal year.

(2) CSL: “Competitive Sustainment Level.” Laboratory equipment that if not funded will result in a laboratory program in AMBER status. If funded this would allow a laboratory program to maintain a GREEN status through the current fiscal year.

(3) FSL: “Full Sustainment Level.” Laboratory equipment needed to support development of enhanced capabilities (margin of excellence) or an optimal scheme of life-cycle replacement.

c. The LERC will recommend to the Dean the allocation of OMA funds to individual departments that have laboratory programs. The process of funds allocation includes determining departmental requirements, prioritizing requirements, and preparing a UFR list for unfunded requirements.

(1) The LERC will meet in September to conduct the funds allocation process. Each department will submit their laboratory program requirements for S&M and R&U items for the upcoming fiscal year in an automated Excel spreadsheet file tailored to each department. The first sheet identifies the laboratory requirements funding for the current fiscal year (example at ENCLOSURE B). On this sheet each department will submit requirements for Supply and Maintenance (S&M) to support all laboratory programs in that department. These funds are used to purchase consumable supplies and very small amounts of non-consumable supplies (less than 10% of a lab program’s S&M budget) to support laboratory programs. In addition, these funds support regular and programmed maintenance (not warranties) on equipment supported by the LERC. This spreadsheet will contain a summary of the status of each laboratory program (Green/Amber/Red/Black) and a total funding requirement by equipment classification (MSL/CSL/FSL).

(2) As part of the funds allocation process each department also submits a list of R&U laboratory equipment items. The second sheet in the Excel file is used to identify a prioritized listing of equipment requirements by MSL/CSL/FSL classification (example at ENCLOSURE C). A laboratory equipment item is a piece of equipment, an integrated system composed of different pieces of equipment, or multiple identical pieces of equipment purchased from a single vendor. A laboratory equipment item may not be multiple separate pieces of equipment from a single vendor or multiple vendors to support a laboratory. The department determines which laboratory program is supported by that laboratory equipment item. In addition, each laboratory equipment item is categorized in one of the following R&U categories: increased capability, life-cycle replacement, equipment upgrade, technology replacement, and new capability. These

categories may be used in the process of prioritizing requirements (See Para. 7 Requirements Resourcing Methodology).

(3) The LERC will consider all S&M and laboratory equipment requirements submitted by the departments to prioritize these requirements among the departments (See Para. 7 Requirements Resourcing Methodology). The top priority for funding will be for S&M and MSL items, while second and third priority will be for funding CSL and FSL items, respectively. This process will likely require the LERC to distinguish the priority of items among departments within the same category (MSL, CSL, FSL). If there are sufficient funds to fully resource S&M and MSL requirements, the Chair of the LERC will recommend to the committee to fully fund all S&M and MSL items. If there are insufficient funds to fully resource all S&M and MSL items recommended by the departments, the LERC will then create one prioritized list of recommended MSL items and allocate funding to the highest priority items across the departments. If, after funding all S&M and MSL items sufficient funds remain to resource all CSL items, the Chair of the LERC will recommend to the committee to fully fund all CSL items. If after funding of S&M and MSL items, there are insufficient funds remaining to fund all CSL items, the LERC will then create one prioritized list of recommended CSL items and allocate funding to the highest priority items across the departments. FSL items will not be funded ahead of MSL or CSL items.

(4) During the funding process the LERC will also allocate sufficient funds to support an Insurance Account available for use by all laboratory programs. This account is used to fund the unanticipated repair of laboratory equipment that is considered vital to a laboratory program (See Para. 7 Requirements Resourcing Methodology). Any funds remaining in this account towards the end of the fiscal year will be used to purchase high priority equipment on the UFR list.

(5) All unfunded laboratory equipment items will be included in a UFR list that is submitted to the VDR (ENCLOSURE D). The UFR list will be updated throughout the year as items are added or deleted from it. To differentiate UFR items from regular requisitions, LERC departments will annotate "UFR" in the upper right-hand corner of the request when pre-positioning them at the Directorate of Resource Management.

(6) The total funds allocated to the departments and the Insurance account for S&M and laboratory equipment items are detailed in the Laboratory Funding Allocation table (ENCLOSURE E). The table also indicates the total number of laboratory programs in GREEN, AMBER, RED, and BLACK status for each department because of the recommended funding allocations.

(7) The laboratory Funding Allocations table is submitted through the Assistant Dean of the Plans and Resources Division (PRD) and the VDR to the Dean for approval.

6. Procurement Procedures. Once funds have been allocated by the LERC and approved by the Dean, equipment and supplies may be purchased by the Departments using the legal, proper, and correct procedures.

7. Requirements Resourcing Methodology. The preceding sections have described the funds allocation. The following sections identify a methodology for funding resource requirements within the constraints of this system.

a. Supply and Maintenance. The priority for funding is to sustain the laboratory equipment on hand. These elements are funded at the beginning of the fiscal year and are not appropriate for UFRs.

(1) Annual Supply Costs. These costs include supplies directly associated with a particular piece of laboratory equipment or with projects that are constructed or tested in the laboratory. Examples of equipment affiliated supply costs are air filters, gaskets, grommets, plotter pens, laser print cartridges, and recordable disks. Examples of project affiliated supply costs are chemical reagents, consumable electronic circuit components, and concrete casting forms.

(2) Maintenance Costs.

(a) Maintenance costs include the cost of calibration or other periodic maintenance services that are required to maintain the equipment in a useful condition. [Most calibration should be performed by the U.S. Army Test, Measurement, and Diagnostic Equipment Activity \(USATA\).](#)

(b) Maintenance costs are normally predictable and must be scheduled. Therefore, these requirements must be resourced as financed requirements throughout the fiscal year. Where services must be performed by other than department personnel, it may be necessary to get a service contract that covers the fiscal year.

(3) Repair Costs.

(a) This is the cost of repairing a broken piece of laboratory equipment not under warranty and not on the [contractor/DOIM density list](#).

(b) Only items that must have a maintenance contract will have a repair contract. Normally the repair service is a part of the maintenance agreement. Service contracts are expensive and in general will not be used for repair of lab equipment.

(c) The LERC will allocate funds to an insurance account as an emergency repair fund. When departments have an equipment item or items that fail, they may request allocation of funds from the Insurance account. These requests are made through the Chair of the LERC.

(4) Software/Firmware Upgrade and Maintenance Costs.

(a) Some software packages require maintenance contracts that keep the software supported with patches and service packs, allow users to get technical help or otherwise keep the software functional. These package costs must be scheduled.

(b) Some software maintenance packages have licenses that expire. For software that is a critical part of the laboratory program, this cost must be scheduled.

(c) Some software maintenance packages are required to keep the software operational and secure. It may also keep the software compatible with the USMA computing environment. Maintenance to keep the software compatible with the USMA computing environment must be scheduled even if it adds significant functionality as a side effect.

b. Equipment Replacement and Upgrade. There are circumstances under which new equipment is required for a laboratory. They are listed in the following paragraphs in no order of priority. Departments are responsible for determining the priority of their own academic laboratory requirements and defend them before the LERC. Ultimately, in the context of departmental priorities, the LERC will determine the priority of laboratory requirements for all departments in this category. If funds are available, these requirements will be funded. If inadequate OMA dollars are allocated for LERC to resource these requirements, the equipment needs should be included in the UFR package.

(1) Increased Capacity Requirements. When course enrollments increase, the quantity of equipment required to support the course may also increase. Equipment in this category should not increase the function of the laboratory but increase the number of stations.

(2) Life Cycle Replacement. This category accommodates the need to replace worn out equipment and should be scheduled over a number of years based on an estimate of the life cycle of the equipment.

(3) Equipment Upgrade. Any hardware improvement that increases the functionality of a piece of equipment, short of an actual replacement, falls into this category. Accessories for test and measurement equipment, memory expansion cards, and CD ROMs are examples.

(4) Technology Replacement. This category includes items of equipment that require replacement on a more frequent interval than the wear-out interval because of advances in technology. Computers are good examples of pieces of equipment that are obsolete well before they are worn out. Technology replacement for compatibility, especially meeting security requirements falls under Supply and Maintenance. (Para 7.a.).

(5) New Capability. This category includes every item of requested equipment that is not a replacement for equipment currently at USMA. New Capability Equipment increases the function of the laboratory program.

(6) Program Realignment. This category includes items that are requested because of an approved shift in laboratory program. Resource implications must be considered and approved before such equipment can be included in the LERC budget.

8. Projecting Future Year Requirements.

a. In April of each fiscal year the LERC projects laboratory program requirements for the next fiscal year. Departments will prepare projected requirements using the automated spreadsheet file shown in ENCLOSURE B and C.

b. The total projected laboratory requirements are detailed in the Future Years Laboratory Requirements table (ENCLOSURE F). The Future Years Laboratory Requirements table is submitted through the Assistant Dean PRD to the VDR for review.

c. The LERC will also develop and maintain a list of projected OPA-2 laboratory projects for the next five fiscal years.

9. Requesting the Establishment of a New Laboratory Program. To add a new laboratory program or substantially change a laboratory program, a department will staff a proposal to the other departments and then through the Curriculum Committee to the General Committee for

approval. The Curriculum Committee will review Part a. below to ensure that Dean-funded laboratory programs are aligned with the institutional curriculum. Parts b. and c. will be reviewed by the VDR. The proposal must address the following questions:

a. Academic need for a new laboratory program. (For Curriculum Committee)

(1) What is the academic need for the addition of the new laboratory program? What course(s) in the curriculum will the new laboratory program support?

(2) What Student Learning Outcomes (SLOs) in the course(s) is the new laboratory program tied to? What is the plan for assessing those SLOs in the course(s) tied to the new laboratory program?

(3) What is the creation timeline for the new laboratory program?

b. Equipment needed for the new laboratory program. (For LERC, PRD-B, and VDR)

(1) What equipment is needed to establish the new laboratory program? What is the academic need for the equipment? A prioritized list in an enclosure is required.

(2) What items presently on hand in a department or laboratory program will be "transferred" to the new laboratory program? A designation in the first enclosure or a second enclosure is required.

(3) What equipment exists in other laboratory programs that the new laboratory program could make use of through a sharing agreement instead of making a purchase? An endorsement from the equipment-sharing department is highly encouraged.

c. Physical and financial requirements for the new laboratory program. (For LERC, PRD-B, and VDR)

(1) What physical footprint inside the academic buildings will the new laboratory program occupy? Are there construction costs associated with establishing the new laboratory program space? An endorsement from the VDR is required for increased or modified laboratory space beyond the purchase of equipment.

(2) What is the start-up cost to purchase new equipment needed in the first enclosure? What is the timeline for purchasing the equipment? What organization will be funding the start-up costs of the new laboratory program? This can be the LERC, but the requester should recognize that all other laboratory programs will take a financial hit if the LERC is the funder. Including an endorsement from the source for start-up funding is highly recommended.

(3) Once established, the LERC would supply the S&M and R&U funds. What is the estimated LERC budget for the new laboratory program for the first 5 years of execution?

10. Laboratory Equipment and Resource Committee.

a. Committee Composition. The LERC is composed of 11 voting members. The individual serving as members representing the following departments are recommended by the Head of the respective departments.

396 Behavioral Science and Leadership
397 Chemical and Biological Science and Engineering
398 Civil and Environmental Engineering
399 English and World Languages
400 Electrical Engineering and Computer Science
401 Geography and Earth Sciences
402 History and War Studies
403 Mathematical Sciences
404 Mechanical and Aerospace Engineering
405 Physics and Nuclear Engineering
406 Systems Engineering
407

408 b. The Dean selects a Faculty Council member to serve as Chair of the LERC. The LERC
409 Chair is a nonvoting member and will serve for a term no less than two years. The Chair may
410 not simultaneously serve as a departmental representative.
411

412 c. The following people, or their appointed delegate, are also invited to participate in the
413 LERC as non-voting members:
414

- 415 (1) the Assistant Dean for Programs and Resources,
- 416
- 417 (2) the Associate Dean for Informational Technology,
- 418
- 419 (3) the Chief Information Officer (CIO),
- 420
- 421 (4) a representative from the United States Military Academy Preparatory School, and
- 422
- 423 (5) a representative from each of the G4 and G8 offices.
424

425 d. Faculty and staff representatives from across USMA are welcome to attend the LERC
426 meeting and participate in the budget allocation discussions. These representatives do not have
427 voting rights.
428

429 e. Committee Responsibilities. Members of the committee represent the Faculty Council
430 and the faculty at large in providing the Dean recommendations concerning the allocation of
431 resources for academic laboratories. The committee seeks to enhance the faculty involvement
432 and representation in the resourcing decisions that affect academic laboratories, working closely
433 with the Associate Dean for Informational Technology, the CIO/G8 and the Plans and
434 Resources Division. Members of the committee ensure that their departments understand these
435 procedures and align department resource planning with the provisions of this plan. Specific
436 responsibilities of the committee include the following:
437

- 438 (1) Prepare a proposal of supply and equipment requirements to support laboratory
439 programs in the current fiscal year.
440
- 441 (2) Track the commitments and obligations of departmental laboratory funding.
442
- 443 (3) Maintain a departmental UFR list for mid-year and year-end funding.
444
- 445 (4) Submit laboratory funding requirements for future fiscal years.
446

(5) Maintain a plan for maintenance and replacement of laboratory equipment.

f. The Office of the Dean's Responsibilities. The Office of the Dean will provide administrative and clerical support for the committee. Functions to be performed include the following:

(1) Provide funding guidance to the committee.

(2) Consolidate department UFR submissions.

(3) Archive committee funding allocation decisions.

(4) Provide data of commitments and obligations to the committee as needed.

11. Fiscal Year Schedule.

a. Resource managers in the departments should recognize the general resource cycle.

(1) Once funds are allocated to the department, the department must decide what items will be procured, assuming the funds are less than requested. Budget execution then begins.

(2) Once the execution of the budget is underway, the commitment rate is monitored. At this time the department will also prepare the list of requirements that were not funded by the department allocation for the UFR submission.

(3) In the final month of the fiscal year, procurement actions must be closely monitored to ensure that all deadlines are met.

b. The timing for allocation of OPA funds is unpredictable and can happen at any time during the fiscal year. Whenever funding does occur, action against this funding must begin immediately. While these funds are often multiple year funds, slow commitment rates can result in the Department of the Army withdrawing funds or under resourcing requirements in subsequent years.

12. **Proponent.** The proponent for this plan is the Vice Dean for Operations.

FOR THE DEAN OF THE ACADEMIC BOARD:

KRISTA WATTS
COL, MP
Vice Dean for Operations

ENCLOSURES:

A – Academic Laboratory Programs
B – Laboratory Requirements Table Example
C – Laboratory Equipment Items Table Example
D – Prioritized Unfinanced Requirements Tables
E – Laboratory Funding Allocations Table
F – Future Years Laboratory Requirements Table

497 **Behavioral Science and Leadership**

498 Cognitive Neuroscience

499 Human Factors

500 Leadership Laboratory

501

502 **Chemical and Biological Science and Engineering**

503 Chemical Engineering

504 Chemistry Electives

505 General Chemistry

506 Individual Research

507 **Biology**

508

509 **Civil and Environmental Engineering**

510 Capstone and General Support Shops

511 **Environmental Engineering**512 **Environmental Engineering Sequence**

513 Engineering Mechanics

514 Geotechnical

515 Hydrology and Hydraulics

516 Structural Engineering

517 **Resilience and Sustainability**

518

519 **Electrical Engineering and Computer Sciences**

520 Basic Electronics

521 Computer Science General Purpose Curriculum

522 Computer Science Server and Network Support

523 Computer Science Dedicated Purpose Labs

524 Core Information Technology

525 Digital Systems

526 Electrical Engineering Advanced Elective

527 General Support and Photolithography

528

529 **English and World Languages**530 **Learning Resource Center**531 **Languages Laboratory**532 **Language A/V Studio**

533

534

535 **Geography and Earth Sciences**

536 Core Geography

537 **Environmental Science**

538 Geography Electives

539 **Geospatial Science**

540

541 **History and War Studies**

542 History Laboratory

543

544 **Mathematical Sciences**

545 Math Courses

546

547 **Mechanical and Aerospace Engineering**548 **Capstone and General Support**

549 Engineering Computing

550 **Mechanics, Materials, and Manufacturing**

551 Thermal Sciences

552 **Aerospace Engineering**553 **Dynamical Systems**

554 Mechanical Systems

555

556 **Physics and Nuclear Engineering**557 **General Support**558 **Space Science**

559 Core Physics

560 Experimental Physics

561 Lasers & Optics

562 Nuclear Engineering

563 Radiation Safety

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565 **Systems Engineering**

566 Systems Engineering Majors

567 Systems Engineering Sequence

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570 **Red** = New lab program571 **Green** = Moved to another department572 **Blue** = Name change

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PRIORITIZED UNFINANCED REQUIREMENTS**Date:** _____

Unfunded CSL items: Failure to fund this project will cause the associated laboratory program to stay in an AMBER status. The general impact of not funding these projects is to increase cadet/faculty time on task thus reducing overall efficiency. As additional funds are available through reallocation, the LERC will choose projects from within the highest available priority group.

Item	Cost (\$k)	Laboratory Program	Dept	Course(s) Supported & Remarks

Unfunded FSL items: Funding these projects above the GREEN status is to support the development of enhanced capabilities or an optimal scheme of life cycle replacement. The projects are listed by department in alphabetical order and are not listed by priority. As additional funds are available through reallocation, the LERC will first fund projects from the unfunded CSL project list and then choose from these FSL projects within the highest available priority group.

Item	Cost (\$k)	Laboratory Program	Dept	Course(s) Supported & Remarks

Laboratory Requirements Funding - FY25						
Department Summary						
Date Prepared 8/26/2024						
LERC Budget for FY2025		3100.0 k\$				
Department	Laboratory Program	S&M	Status	Replacement & Upgrade		
		k\$		MSL	CSL	FSL
						total
BS&L	Cognitive Neuroscience	10.0	Green	36.0	0.0	46.0
	Human Factors	30.0	Red	0.0	10.0	40.0
	Leadership Laboratory	18.0	Red	10.0	2.0	30.0
	Department Totals:	58.0		46.0	12.0	116.0
	Department Targets:	58.0		46.0	50.5	154.5
C&LS	Chemical Engineering	75.0	Green	0.0	82.5	157.5
	Chemistry Electives	75.0	Green	0.0	0.0	75.0
	General Chemistry	40.0	Green	0.0	0.0	40.0
	Individual Research	35.0	Green	0.0	0.0	35.0
	Biology	100.0	Amber	0.0	0.0	100.0
		Department Totals:	325.0		0.0	82.5
	Department Targets:	325.0		0.0	262.5	864.5
CME	Capstone and General Support Shops	59.0	Green	75.0	0.0	134.0
	Engineering Computing	42.0	Green	0.0	0.0	42.0
	Engineering Mechanics	20.0	Green	0.0	0.0	20.0
	Geotechnical	10.0	Green	0.0	0.0	10.0
	Hydraulics and Hydrology	10.0	Green	0.0	0.0	10.0
	Mechanical Systems	42.0	Green	0.0	0.0	42.0
	Structural Engineering	15.0	Green	0.0	25.0	40.0
	Thermal Sciences	132.0	Green	185.0	0.0	317.0
		Department Totals:	330.0		260.0	25.0
	Department Targets:	298.0		260.0	395.0	1518.0
DEWL	Audio Lab (LRC)		Amber	0.0	0.0	0.0
	Language Recording Studio (LRS)	4.0	Amber	0.0	2.0	6.0
	Virtual Immersion Competitive-Cooperative Space (VICCS)	26.0	Green	6.2	0.0	32.2
	Department Totals:	30.0		6.2	2.0	38.2
	Department Targets:	19.0		6.2	2.0	102.2
EECS	Basic Electronics	0.0	Green	0.0	0.0	0.0
	Comp Sci Gen Purp Curriculum	0.0	Green	0.0	0.0	0.0
	Comp Sci Server & Network Suppt	0.0	Red	0.0	0.0	0.0
	Computer Sci Dedicated Purp Labs	0.0	Green	0.0	0.0	0.0
	Core Information Technology	0.0	Red	0.0	0.0	0.0
	Digital Systems	0.0	Green	0.0	0.0	0.0
	Electrical Eng Advanced Elective	0.0	Amber	0.0	0.0	0.0
	General Support & Photolithography	0.0	Amber	0.0	0.0	0.0
	Department Totals:	320.0		0.0	0.0	320.0
	Department Targets:	320.0		0.0	268.0	588.0
GEnE	Core Geography	8.4	Green	0.0	0.0	8.4
	Environmental Engineering Sequence	13.2	Red	525.1	0.0	538.3
	Env Science and Engineering	37.0	Red	140.0	0.0	177.0
	Geography Electives	6.0	Green	0.0	0.0	6.0
	Geospatial Information Science	84.0	Red	17.0	58.0	159.0
	Department Totals:	148.6		682.1	58.0	888.7
	Department Targets:	148.6		682.1	58.0	1405.7
History	Digital History Center	7.0	Green	0.0	2.4	9.4
	Department Totals:	7.0		0.0	2.4	9.4
	Department Targets:	7.0		0.0	2.4	9.4
Math	Math Courses	2.0	Red	7.3	0.0	9.3
	Department Totals:	2.0		7.3	0.0	9.3
	Department Targets:	2.0		7.3	0.0	9.3
P&NE	General Support	5.0	Red	75.0	0.0	80.0
	Space Science	16.1	Amber	0.0	0.0	16.1
	Core Physics	20.0	Amber	0.0	23.5	43.5
	Experimental Physics	16.7	Green	0.0	0.0	16.7
	Lasers & Optics	7.9	Amber	125.0	0.0	132.9
	Nuclear Engineering	23.0	Red	0.0	0.0	23.0
	Radiation Safety	3.7	Green	0.0	0.0	3.7
	Department Totals:	92.4		200.0	23.5	315.9
	Department Targets:	103.0		200.0	238.7	580.3
SE	Systems Engineering Electives	29.0	Red	207.8	0.0	236.8
	Systems Engineering Sequence	43.0	Green	0.0	0.0	43.0
	Department Totals:	72.0		207.8	0.0	279.8
	Department Targets:	72.0		207.8	76.8	356.6
LERC	Emergency Fund	100.0	Continuation of the policy from last year			
S&M total includes Emergency Fund		S&M	Status	Replacement & Upgrade		
		k\$		MSL	CSL	FSL
						Total
Totals (Allocated 2025)		1485.0		1409.4	205.4	3099.8
Targets (Requested 2025)		1482.9		1409.4	1353.9	6260.5
Unfunded				0.0	1148.5	3160.7
# Lab Programs		11	Red			
		8	Amber			
		24	Green			
		43	total lab programs			

Future Years Laboratory Requirements

Date prepared:

In thousands of dollars.

Department	FY+1				FY+2		FY+3		FY+4		FY+5		FY+6	
	S&M	R&U			S&M	R&U	S&M	R&U	S&M	R&U	S&M	R&U	S&M	R&U
		MSL	CSL	FSL										
BS&L														
CBSE														
CEE														
EE&CS														
DEWL														
GEO														
HWS														
Math														
MAE														
P&NE														
Systems														
Total														